No. 2

Date 97 · 7 · 8

H-like ion の イオン化 エネルギー $\chi_I = 13.6\,eV\,Z^2$ $\chi_I = 1 \sim 10\,keV$

$\Rightarrow Z = 8 \sim 27$ $\Rightarrow O \sim Fe$

J.N.Bahcall & C.L. Sarazin ApJ 219 (1978) 781

	λ(Å)	E(eV)
O	19.0	0.7
Ne	12.2	1.0
Mg	8.4	1.5
Si	6.2	2.0
Ca	3.0	4.1
Fe	1.8	6.9

(ii) continuum $\Rightarrow$ thermal bremsstrahlung. G.B. Rybicki & A.P. Lightmann (J.Wiley 1979)

"Radiative process in Astrophysics"

Maxwell 分布 $f(v) = \left(\frac{m_e}{2\pi kT}\right)^{3/2} 4\pi v^2 \exp\left(-\frac{m_e v^2}{kT}\right)$ (1.2.3)

をしている電子のだす制動輻射

@ emissivity [erg/s/cm³/Hz]

$$\frac{d^2 L_X}{dV d\nu} = \frac{2^5 \pi e^6}{3 m_e c^3}\left(\frac{2\pi}{3 m_e kT}\right)^{1/2} \times n_e \sum_i Z_i^2 n_i e^{-\frac{h\nu}{kT}} \bar{g}_{ff}(T,\nu) \quad (1.2.4)$$

@ bolometric luminosity per unit volume [erg/s/cm³]

Gaunt 因子 $\sim O(1) \simeq 0.9\left(\frac{h\nu}{kT}\right)^{-0.3}$

$$\frac{dL_X}{dV} = \left(\frac{2\pi kT}{3 m_e}\right)^{1/2}\left(\frac{2^4 e^6}{3\pi m_e c^3}\right) n_e \sum_i Z_i^2 n_i \bar{g}_{ff}(T) \quad (1.2.5)$$

$$\sim 5 \times 10^{-24}\left(\frac{T}{1keV}\right)^{1/2}\left(\frac{n_e}{1cm^{-3}}\right)^2 erg/s/cm^3 \quad (1.2.6)$$

@ typical luminosity

$$L_X \sim V\frac{dL_X}{dV} \sim 10^{43}\,\xi\left(\frac{\Omega_b}{0.1}\right)^2\left(\frac{T}{1keV}\right)^{1/2}\left(\frac{M_{cl}}{10^{14}M_\odot}\right)^{-2}\left(\frac{1Mpc}{R_{cl}}\right)^3 [erg/s] \quad (1.2.7)$$

ξ clumping factor

下限！

$$V_{cl} \sim R_{cl}^3 \sim (3\times10^{24})^3\left(\frac{R_{cl}}{1Mpc}\right)^3$$

$$n_e \sim \frac{\Omega_b M_{cl}}{m_p V} \sim 3\times10^{-3}\Omega_b\left(\frac{M_{cl}}{10^{14}M_\odot}\right)\left(\frac{1Mpc}{R_{cl}}\right)^3$$

1.3 Why X-ray ?

※ surface brightness

(i) optical

銀河団 = 銀河の重ね合わせ $\propto \int n_{gal}\,dl$

$$\left(\frac{S_{cluster}}{S_{background}}\right)_{opt} = \frac{\int_{-R_{cl}}^{R_{cl}} n\,dl}{\int_0^\infty n\,dl} \sim \frac{\langle n_{gal}\rangle_{cl} \times 2R_{cl}}{\langle n_{gal}\rangle_{univ} \times c/H_0} \quad (1.3.1)$$

$\langle n_{gal}\rangle_{univ} \sim 0.01/(h^{-1}Mpc)^3$ (1.3.2)

$\langle n_{gal}\rangle_{cl} \sim \frac{100}{\frac{4\pi}{3}R_{cl}^3} \sim \frac{100}{4\times(1.5)^3} \sim 10/(h^{-1}Mpc)^3$ (1.3.3)

$$\rightarrow \left(\frac{S_{cl}}{S_{back}}\right)_{opt} \sim 10^3 \times \frac{3h^{-1}Mpc}{3000\,h^{-1}Mpc} \sim 1 \quad (1.3.4)$$

optical で cluster を同定することは極めて困難

$\Rightarrow$ projection contamination $\Rightarrow$ 2D → 3D obs

KOKUYO

COLLEGE NOTES / n028 Source p.2 / 1 of 15 selected pages

No. 3

Date 97 · 7 · 8

(iii) X-ray

$S_X \propto \int n_e^2 dl \propto \int n_{gal}^2 dl$

⇒ X-ray selected

$$\left(\frac{S_{cl}}{S_{back}}\right)_{XR} \sim \frac{\langle n_{gal}^2\rangle_{cl}}{\langle n_{gal}^2\rangle_{univ}} \times \frac{2R_{cl}}{c/H_0} \gtrsim \frac{\langle n_{gal}^2\rangle_{cl}}{\langle n_{gal}^2\rangle_{univ}} \times 10^{-3} \sim 10^3 \quad (1.3.5)$$

§2 Gas distribution in clusters

2.1 hydrostatic equilibrium : isothermal β-model

cluster = gas + galaxies (+DM) が球対称で平衡状態にある

● gas 静水圧平衡

$$\frac{1}{\rho_{gas}}\frac{\partial P_{gas}}{\partial r} = -\frac{GM}{r^2} \quad (2.1.1) \qquad \leftarrow \text{total} : \text{gas} + \text{gal} + \text{DM}$$

$$P_{gas}(r) = n_{gas}(r)\, k\, T_{gas}(r) = \frac{\rho_{gas}}{\mu m_p} kT_{gas} \quad (2.1.2)$$

★ mean molecular weight 完全に電離した H と He $\quad X \equiv \frac{n_H}{n_H + 4n_{He}}$, $Y = \frac{4n_{He}}{n_H+4n_{He}} = 1-X$

$$(n_e + n_H + n_{He})\mu m_p = n_H m_H + n_{He} m_{He} \qquad \mu = \frac{4}{5X+3} \quad (2.1.5) \sim 0.59$$

$X = 0.75$

(2.1.2) を (2.1.1) に代入

$$M(r)_{total} = -\frac{r^2}{G\rho_{gas}}\left[\frac{dn_{gas}}{dr} kT_{gas} + n_{gas}\frac{d(kT_{gas})}{dr}\right] = -\frac{kT_{gas}\, r}{G\mu m_p}\left[\frac{d\ln n_{gas}}{d\ln r} + \frac{d\ln T_{gas}}{d\ln r}\right] \quad (2.1.6)$$

★ 銀河 : Jeans eq. $\quad \frac{1}{\rho_{gal}}\frac{\partial}{\partial r}\left[\rho_{gal}\sigma_r^2\right] + \frac{2}{r}(\sigma_r^2 - \sigma_t^2) = -\frac{GM(r)}{r^2} \quad (2.1.7)$

$M(r)$ を (2.1.6) と (2.1.7) から消去 $\rho_{gal} \longleftrightarrow \rho_{gas}$, さらに等温を仮定 $\quad A \equiv 1 - \frac{\sigma_t^2}{\sigma_r^2} \quad (2.1.8)$

anisotropy parameter

$$\frac{kT_{gas}}{\rho_{gas}}\frac{\partial n_{gas}}{\partial r} = \frac{\sigma_r^2}{\rho_{gal}}\frac{\partial \rho_{gal}}{\partial r} + \frac{2}{r}\sigma_r^2 A$$

$$\frac{\partial \ln n_{gas}}{\partial r} = \underbrace{\frac{\mu m_p \sigma_r^2}{kT_{gas}}}_{\beta\ (2.1.10)}\left[\frac{\partial \ln \rho_{gal}}{\partial r} + \frac{2}{r}A\right] \quad (2.1.9) \Rightarrow n_{gas}(r) = n_0\left[\frac{\rho_{gal}(r)\, r^{2A}}{\rho_0\, r_0^{2A}}\right]^{\beta} \quad (2.1.11)$$

通常は

isotropic orbit $A=0$ から King model $\quad \rho_{gal}(r) = \rho_0\left[1+\left(\frac{r}{r_0}\right)^2\right]^{-\frac{3}{2}} \quad (2.1.13)$

empirical

$$n_{gas}(r) = n_0\left[1+\left(\frac{r}{r_0}\right)^2\right]^{-\frac{3}{2}\beta} \quad (2.1.14)$$

isothermal β-model $\quad r_0$: core radius

KOKUYO

No. 4

Date 97 · 7 · 8

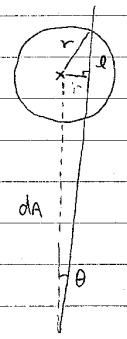
2.2 β - problem

✱ $\beta_{spec} \equiv \frac{\mu m_p \sigma_r^2}{kT_{gas}}$ ← 銀河の速度分散 (optical)

← X-ray

✱ β_{fit} (β_{image})

X-ray surface brightness から決まる $\beta_{fit} = \beta_{spec}$??



$$S_X(\theta) \propto \int_{-\infty}^{\infty} dl\, n_{gas}^2(r=\sqrt{l^2+d_A^2\theta^2}) \propto \int_{-\infty}^{\infty} \left[1+\frac{l^2+d_A^2\theta^2}{r_0^2}\right]^{-3\beta} dl$$

$$= \left[1+\frac{\theta^2}{(r_0/d_A)^2}\right]^{-3\beta} \int_{-\infty}^{\infty} \left[1+\frac{l^2}{r_0^2+d_A^2\theta^2}\right]^{-3\beta} dl \qquad \left(\frac{r_0}{d_A}=\theta_0\right)$$

$$= \left[1+\left(\frac{\theta}{\theta_0}\right)^2\right]^{-3\beta} \cdot 2\sqrt{r_0^2+d_A^2\theta^2} \int_0^{\infty} \frac{dx}{(1+x^2)^{3\beta}} = \sqrt{\pi}\, r_0 \left[1+\left(\frac{\theta}{\theta_0}\right)^2\right]^{-3\beta+\frac{1}{2}} \frac{\Gamma(3\beta-\frac{1}{2})}{\Gamma(3\beta)} \qquad (2.2.1)$$

$$\frac{\Gamma(3\beta-1/2)\sqrt{\pi}}{2\Gamma(3\beta)}$$

$$S_X(\theta) \propto \left[1+(\theta/\theta_0)^2\right]^{-3\beta+\frac{1}{2}}$$

$$\theta_0 = r_0 / d_A(\Omega_0, \lambda_0, z)$$

$\langle \beta_{spec} \rangle = 0.8 \sim 1.2$

#?

$\langle \beta_{fit} \rangle = 0.5 \sim 0.7$

✓ the King model for galaxy ? $\left[1+\left(\frac{r}{r_0}\right)^2\right]^{-\frac{3}{2}p}$ $p=1$ $\beta_{spec} \cdot p = \beta_{fit}$ $p=0.6$

✓ isothermal

✓ cluster 中の galaxy の orbit isotropic A $\begin{cases} (2A-3)\beta_{spec} = -3\beta_{fit} \\ A \sim 0.5 \Rightarrow \sigma_r^2 = 2\sigma_t^2 \end{cases}$

✓ non-hydrostatic equilibrium

2.3 Gas mass fraction

もし cluster 全体で baryon / total mass が $\frac{M_b}{M_{tot}} = \frac{M_{gas}+M_{star}}{M_{tot}} = \frac{\Omega_b}{\Omega_0}$ (BBN) であるとすると

Ω_0 を推定

Ω_b 〃

(i) gas mass

thermal bremsstrahlung
T_{gas}
surface brightness
} $n_e(r)$

$$\rho_{gas}(r) = m_H \underbrace{n_H(r)}_{\frac{2X}{1+X} n_e} + m_{He} \underbrace{n_{He}(r)}_{\frac{1-X}{2(1+X)} n_e} = \frac{2}{1+X} m_p n_e(r) \qquad (2.3.4)$$

(2.3.3)

KOKUYO

$$M_{gas}(<r) = \int_0^r \rho_{gas}(r)\, 4\pi r^2 dr = \frac{8\pi m_p}{1+X} n_{e0} \int_0^r \left[1+\left(\frac{r}{r_0}\right)^2\right]^{-\frac{3}{2}\beta} r^2 dr = \frac{8\pi m_p}{1-X} n_{e0} r_0^3 \underbrace{\int_0^{r/r_0} (1+x^2)^{-\frac{3}{2}\beta} x^2 dx}_{I_M(r/r_0,\beta)} \quad (2.3.5)$$

$$I_M(x,\beta) = \frac{1}{2}\left[B\left(\tfrac{3}{2}\beta-\tfrac{3}{2},\tfrac{3}{2}\right) - B_{\frac{1}{1+x^2}}\left(\tfrac{3}{2}\beta-\tfrac{3}{2},\tfrac{3}{2}\right)\right]$$

$$B(p,q) = \int_0^1 t^p (1-t)^{q-1} dt \quad (2.3.6)$$

$$B_z(p,q) = \int_0^z \quad "$$

(ii) total mass

optical

$$M(r) = -\frac{r^2}{G}\left[\frac{1}{\rho_{gal}}\frac{d(\rho_{gal}\sigma_r^2)}{dr} + 2\frac{\sigma_r - \sigma_t}{r}\right] \quad (2.3.7)$$

X-ray

$$M(r) = -\frac{kT_{gas}r}{G\mu m_p}\left[\frac{d\ln n_{gas}}{d\ln r} + \frac{d\ln T_{gas}}{d\ln r}\right] \quad (2.3.8)$$

2.4 Ω_0 from coma cluster

S.D.M. White etal Nature 366 (1993) 429

optical

$$M_{tot}(<1.5\,h^{-1}\mathrm{Mpc}) = (6.7\pm1.0)\times10^{14}\,h^{-1}M_\odot \quad (2.4.1)$$

X-ray

$$" \quad = (6.82\pm0.34) \quad " \quad (2.4.2)$$

• gas $M_{gas}(<1.5\,h^{-1}\mathrm{Mpc}) = (5.45\pm0.98)\times10^{13}\,h^{-5/2}M_\odot \quad (2.4.3)$

• stars in galaxies

$$L_B(<1.5\,h^{-1}\mathrm{Mpc}) = 1.95\times10^{12}\,h^{-2}L_\odot \quad (2.4.4)$$

$$M/L = 8.0\,h\,(L/10^{10}\,h^{-1}L_\odot)^{0.35} \Rightarrow M_{stars}(<1.5\,h^{-1}\mathrm{Mpc}) = (1\pm0.2)\times10^{13}\,h^{-1}M_\odot \quad (2.4.5)$$

$\rightarrow 6.4\,h$

• baryon fraction

$$F_b(<1.5\,h^{-1}\mathrm{Mpc}) \simeq \frac{M_{gas}+M_{star}}{M_{tot}} = 0.08\,h^{-3/2} + 0.015 \quad (2.4.6)$$

$$\Omega_b = (0.015\pm0.01)\,h^{-2}$$

$$\Omega_0 = \frac{(0.015+0.01)\,h^{-2}}{0.015+0.08\,h^{-3/2}} = \frac{1\pm0.6}{h^2+5h^{-3/2}} \simeq (0.16\pm0.1) \quad (2.4.7)$$

$h\sim0.7$

§3 The Sunyaev – Zel'dovich effect

refs Zel'dovich & Sunyaev Astrophys. Sp. Sci 4 (1969) 301

Sunyaev & Zel'dovich " 7 (1970) 20

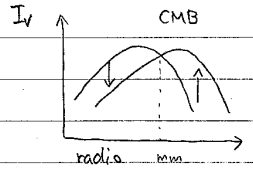
Sunyaev : Sov. Astron Lett 6 (1980) 213

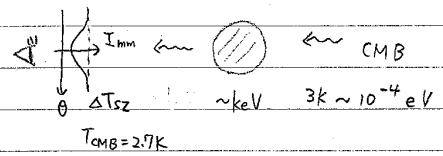
COLLEGE NOTES / n028
Source p.5 / 4 of 15 selected pages

No. 6

Date 97・7・8

3.1 Inverse Compton Scattering of the cosmic microwave background (CMB)





銀河団

radio "negative" source

sub mm "positive" source ?

* Kompaneets 方程式

温度 T_e を持つ非相対論的プラズマ ($m_e c^2 \gg kT_e$) の分布関数の時間発展を記述

$$t_c \frac{\partial f}{\partial t} = \left(\frac{kT_e}{m_e c^2}\right) \frac{1}{x_e^2} \frac{\partial}{\partial x_e} \left\{ x_e^4 \left[\frac{\partial f}{\partial x_e} + f(f+1) \right] \right\} \quad (3.1.1)$$

$$\begin{cases} t_c = \dfrac{1}{n_e \sigma_T c} & \text{mean collision time} \quad (3.1.2) \quad = \left(\dfrac{10^{-4}}{n_e}\right) 5 \times 10^{17} \text{ sec} \quad (3.1.5) \\ x_e = h\nu / kT_e & (3.1.3) \end{cases}$$

Bose-Einstein 分布 $f_{BE}(\nu) = \left[\exp\left(\dfrac{h\nu + \mu}{T}\right) - 1 \right]^{-1}$ (3.1.4) $\qquad \dfrac{\partial f_{BE}}{\partial x_e} = -f_{BE}(f_{BE}+1)$

$$t_{cross} \simeq \frac{2R_{cl}}{c} = 3 \times 10^{14} \text{ sec} \quad (3.1.6)$$

$t_c \gg t_{cross}$ optically thin

入射光子は高い精度で Plank 分布 $f_{BB} = \left[\exp\left(\dfrac{h\nu}{kT_{CMB}}\right) - 1 \right]^{-1}$ (3.1.7)

$$f \simeq f_{BB} \qquad \frac{\partial f}{\partial x} \simeq -f(f+1) \quad (3.1.8)$$

$$x \equiv \frac{h\nu}{kT_{CMB}} \quad (3.1.9)$$

(3.1.1) を書き直す

$$t_c \frac{\partial f}{\partial t} = \left(\frac{kT_e}{m_e c^2}\right) \frac{1}{x^2} \frac{\partial}{\partial x} \left\{ x^4 \left[\frac{\partial f}{\partial x} - \frac{T_{CMB}}{T_e} \frac{\partial f}{\partial x} \right] \right\} = \frac{k(T_e - T_{CMB})}{m_e c^2} \frac{1}{x^2} \frac{\partial}{\partial x} \left(x^4 \frac{\partial f}{\partial x} \right) \quad (3.1.10)$$

Compton y-parameter

$$y \equiv \int_0^t \frac{k(T_e - T_{CMB})}{m_e c^2} \frac{dt}{t_c} \simeq \int_0^t \frac{kT_e}{m_e c^2} \sigma_T n_e c \, dt \quad (3.1.11) \qquad \frac{\partial f}{\partial y} = \frac{1}{x^2} \frac{\partial}{\partial x} \left(x^4 \frac{\partial f}{\partial x} \right) \quad (3.1.12)$$

3.2 Spectral distribution due to the SZ effect

$$y \sim \frac{10 \text{keV}}{500 \text{keV}} \times \frac{t_{cross}}{t_c} = 10^{-5} \ll 1 \quad (3.2.1)$$

=> (3.1.12) を摂動で解けばよい！

KOKUYO

$$f(x,y) \simeq f(x,0) + \frac{y}{x^2}\frac{\partial}{\partial x}\left[x^4 \frac{\partial}{\partial x} f(x,0)\right] \quad (3.2.2) \qquad f(x,0) = f_{BB}(x) = \frac{1}{e^x-1}$$

$$= \frac{1}{e^x-1} + \frac{y}{x^2}\left[x^4 \frac{e^x(e^x+1)}{(e^x-1)^2} - 4x^3 \frac{e^x}{(e^x-1)^2}\right] = \frac{1}{e^x-1} + \frac{xye^x}{(e^x-1)^2}\left[x\coth\left(\frac{x}{2}\right) - 4\right] \quad (3.2.3)$$

◎ brightness temperature

$$f(x,y) \equiv \left[\exp\left(\frac{h\nu}{kT_B}\right) - 1\right]^{-1} \quad \Rightarrow \quad T_B(x,y) = \frac{xT_{CMB}}{\ln\left[f^{-1}(x,y)+1\right]} \quad (3.2.4)$$

(3.2.3) と $f = \frac{1}{e^x-1}[1+\alpha(x)]$

↓ (3.2.4)

$$\ln(f^{-1}+1) \simeq \ln\left[(e^x-1)(1-y\alpha)+1\right] = \ln\left[e^x - y\alpha(e^x-1)\right] \simeq x - \frac{y\alpha}{e^x}(e^x-1) \qquad \alpha = \frac{xe^x}{e^x-1}\left[x\coth\left(\frac{x}{2}\right)-4\right]$$

$$\therefore \frac{T_B(x,y)}{T_{CMB}} = 1 + y\left(x\coth\frac{x}{2} - 4\right) \quad (3.2.5)$$

$$\frac{\Delta T_B}{T_{CMB}} = y\left(x\coth\frac{x}{2} - 4\right) \quad (3.2.6)$$

◎ spectral intensity

$$I_\nu(x,y) = \frac{1}{4\pi} \times 2 \times \frac{h\nu}{c^2} 4\pi\nu^2 f(x,y) = \underbrace{\frac{2(kT_{CMB})^3}{(hc)^2}}_{i_0} x^3 f(x,y) = i_0 x^3 f_{BB}(x) + i_0 \frac{x^4e^x}{(e^x-1)^2}\left(x\coth\frac{x}{2} - 4\right)y \quad (3.2.7)$$

$$i_0 = 2.6\times10^{-15}\ \mathrm{erg/s/cm^2/Hz/str}$$

$$\frac{\Delta I_\nu}{I_\nu^{BB}} = y\frac{xe^x}{e^x-1}\left[x\coth\left(\frac{x}{2}\right)-4\right] \quad (3.2.8)$$

◎ fractional Rayleigh-Jeans temperature　R-J 領域の近似　$\Delta I_\nu = \frac{2\nu^2}{c^2} k \Delta T^{RJ}$

$$\frac{\Delta T^{RJ}}{T_{CMB}} = y\frac{x^2e^x}{(e^x-1)^2}\left(x\coth\frac{x}{2} - 4\right) = y\,t(x) \quad (3.2.10) \qquad x = \frac{h\nu_{obs}}{kT_{CMB,now}} \leftarrow z\text{ により}$$

$$\frac{\Delta I_\nu}{I_\nu^{BB}} = y\,S(x) \qquad x = 11.8\left(\frac{0.45\,mm}{\lambda}\right)$$

$$\Delta I_\nu = i_0\, y\, g(x) \qquad (x = 3.83,\ \nu = 218\,GHz,\ \lambda = 1.38\,mm)$$

$$\frac{\Delta I}{I} = \frac{\Delta T}{T} \longrightarrow -2y$$

3.3 Profile of y-parameter

$$n_e(r) = n_0\left[1 + \left(\frac{r}{r_0}\right)^2\right]^{-\frac{3}{2}\beta} \quad (3.3.1)$$

$$y(\theta) = \int_{-\infty}^{\infty} dl\, \frac{kT_e}{m_ec^2}\sigma_T n_e(r) \quad (3.3.2) = \sqrt{\pi}\,\frac{\Gamma\left(\frac{3}{2}\beta - \frac{1}{2}\right)}{\Gamma\left(\frac{3}{2}\beta\right)}\left(\frac{kT_e}{m_ec^2}\right)\sigma_T n_0 r_0\left[1 + \left(\frac{\theta}{r_0/d_A}\right)^2\right]^{-\frac{3}{2}\beta + \frac{1}{2}} \quad (3.3.3)$$

KOKUYO

No. 8

Date 97 · 7 · 8

◎ 典型的な値

$\left.\begin{array}{l}\beta\sim 1,\ T_e\sim 5\,\mathrm{keV},\ n_0\sim 3\times10^{-3}\,\mathrm{cm}^{-3}\\ r_0\sim 0.3\,\mathrm{Mpc}\quad d_A(z=0.1)\sim 300\,\mathrm{Mpc}\end{array}\right\}\quad \frac{r_0}{d_A}\sim 3'$

$\sqrt{\pi}\,\frac{\Gamma(1)}{\Gamma(3/2)}\times10^{-2}\times6.17\times10^{-25}\times3\times10^{-3}\times0.3\times3\times10^{24}$

$=4\times10^{-5}$

$$y(\theta)=4\times10^{-5}\left[1+\left(\frac{\theta}{3'}\right)^2\right]^{-1}\qquad (3.3.4)$$

◎ RJ limit $\quad \frac{\Delta I_\nu}{I_\nu^{BB}}=\frac{\Delta T^{RJ}}{T_{CMB}}=\frac{\Delta T_0}{T_{CMB}}=-2y\qquad (3.3.5)$

◎ Submm $\left(\begin{array}{l}\lambda=0.85\,\mathrm{mm}\quad \nu=350\,\mathrm{GHz}\\ x=6.2\end{array}\right)\quad \frac{\Delta I_\nu}{I_\nu^{BB}}\simeq 14y\sim 6\times10^{-4}\left[1+\left(\frac{\theta}{3'}\right)^2\right]^{-1}$

$$\Delta I_\nu\simeq 710\,y\sim 80\,\mathrm{mJy/(arcmin)^2}\left[1+\left(\frac{\theta}{3'}\right)\right]^{-1}\qquad (3.3.7)$$

cf. submm background Puget et al AA 308 (1996) L5

$\nu B_\nu\simeq 3.4\times10^{-9}(\lambda/0.4\,\mathrm{mm})^{-3}\,\mathrm{W/m^2/str}$

$\lambda=0.85\,\mathrm{mm}$

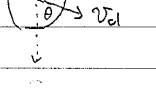
$B_\nu\simeq 110\,\mathrm{mJy/(arcmin)^2}\qquad (3.3.8)\ \ll I_\nu^{BB}$

3.4 Kinematic SZ effect

↑ cf thermal SZ effect

静止している

CMB — Doppler 効果 — Lorentz 変換 $\quad \frac{\Delta T_K}{T_{CMB}}\sim v\cos\theta\qquad (3.4.8)$



$$\frac{\Delta T_K}{T_{CMB}}\simeq v\cos\theta\times\tau=\frac{v_\parallel}{c}\int_{-\infty}^{\infty}n_e\sigma_T\,dl=\left(\frac{v_\parallel}{c}\right)\left(\frac{m_ec^2}{kT_e}\right)y$$

(optical depth)

$$\simeq\underbrace{\frac{300\,\mathrm{km/s}}{3\times10^5}}_{0.1}\times10^2\times\left(\frac{v_\parallel}{300\,\mathrm{km/s}}\right)\left(\frac{5\,\mathrm{keV}}{T_e}\right)y\qquad (3.4.9)$$

thermal SZ：定符号 for given ν

kinematic SZ：± typical ~0.1×(thermal SZ)

$$I_\nu^K=i_0x^3\left[\exp\left(\frac{x}{1+\Delta T_K/T_{CMB}}\right)-1\right]^{-1}$$

$$\Delta I_\nu^K=i_0\underbrace{\left[x^3\frac{xe^x}{(e^x-1)^2}\right]}_{h(x)}\underbrace{\frac{\Delta T_K}{T_{CMB}}}_{\frac{v_0}{c}\tau}\qquad (3.4.10)$$

$$\frac{\Delta I_\nu^K}{I_\nu^{BB}}=\frac{xe^x}{(e^x-1)}\left(\frac{v_0}{c}\right)\tau\qquad (3.4.11)$$

$$\Delta I_{\nu_1}=i_0y\left[g(x_1)+h(x_1)\underbrace{\left(\frac{v_\parallel}{c}\right)\left(\frac{m_ec^2}{kT_e}\right)}_{0.1}\right]$$

$\gg$

$\nu^*=1.4\,\mathrm{mm}\quad g(x^*)=0$

cluster の peculiar velocity $\quad \frac{dh}{dx}\propto(x\coth\frac{x}{2}-4)=0$

$g(x)$

$$\frac{v_\parallel}{c}=\frac{kT_e}{m_ec^2}\,\frac{g(x_2)\Delta I_1-g(x_1)\Delta I_2}{h(x_1)\Delta I_2-h(x_2)\Delta I_1}$$

KOKUYO

No. 9

Date 97 · 7 · 9

3.5 Cosmological distance

- ☆ comoving distance — RW metric の null geodesic $-dt^2 + a^2 \frac{dx^2}{1-kx^2} = 0$
- ☆ luminosity 〃
- ☆ angular diameter 〃

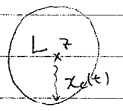
$$\frac{dx}{\sqrt{1-kx^2}} = \frac{dt}{a(t)} \quad (3.5.1)$$

@ luminosity distance

L : absolute bolometric luminosity

F : apparent (observed) luminosity

$$F = \frac{L}{4\pi d_L^2}$$ で d_L を定義



$t = t(z)$ に δt の間隔で isotropic に L を放出、z から $\chi = \chi_c(z)$ で obs

$$L\,\delta t \times \frac{1}{1+z} = F \times 4\pi \left[a_0 \chi_c(z)\right]^2 \times \delta t (1+z)$$

red shift　　dilation

$$F = \frac{L}{4\pi \left[a_0 \chi_c(z)(1+z)^2\right]^2}$$

$$d_L = a_0 \chi_c(z)(1+z) \quad (3.5.8)$$

ただし $\chi \equiv \int \frac{dx}{\sqrt{1-kx^2}} = \begin{cases} \frac{1}{\sqrt{k}} \sin^{-1}\sqrt{k}\,\chi & (k>0) \\ \chi & (k=0) \\ \frac{1}{\sqrt{-k}} \sinh^{-1}\sqrt{-k}\,\chi & (k<0) \end{cases}$

右辺 $\int_0^t \frac{dt}{a(t)} = \frac{1}{H_0} \int_0^z \frac{dz}{\left[\Omega_0(1+z)^3 + (-\Omega_0 - \lambda_0)(1+z)^2 + \lambda_0\right]^{1/2}}$ (3.5.4)

$$K = a_0^2 H_0^2 (\Omega_0 + \lambda_0 - 1) \quad (3.5.5)$$

(i) $K = 0$　$\lambda_0 = 1 - \Omega_0$

$$\chi_c(z) = \frac{1}{a_0 H_0} \int_1^{1+z} \frac{dw}{\sqrt{\Omega_0 w^3 + 1 - \Omega_0}} \quad (3.5.6)$$

(ii) $\lambda_0 = 0$　$\Omega_0 < 1 \to K < 0$

$$\chi_c(z) = \frac{2}{a_0 H_0 \Omega_0^2 (1+z)} \left[2 - \Omega_0 + \Omega_0 z - (2-\Omega_0)\sqrt{1+\Omega_0 z}\right] \quad (3.5.7)$$

Mattig の公式

$$\chi_c(z) \sim \frac{z}{a_0 H_0} \left[1 - \frac{q_0+1}{2} z + O(z^2)\right]$$

$q_0 = \frac{\Omega_0}{2} - \lambda_0$:

@ angular diameter distance

D ↕ θ →　　$\theta \times d_A = D$

$$d_A = a(z)\chi_c(z) = \frac{a_0}{1+z}\chi_c(z) = \frac{d_L}{(1+z)^2} \quad (3.5.9)$$

3.6 Angular diameter distance from the SZ effect

obs : X-ray の surface brightness
　　radio の SZ　　} → d_A → H_0, Ω_0, λ_0
　　z, T_e

(i) X-ray surface brightness

$\nu_1 < \nu < \nu_2$ band [erg/s/cm³]

$$S_x(\theta; \nu_1, \nu_2) = \frac{1}{4\pi(1+z)^4} \int_{-\infty}^{\infty} dl \int_{\nu_1(1+z)}^{\nu_2(1+z)} \frac{d^2 L_x}{d\nu\, dV} d\nu \quad (3.6.1)$$

$$\int_{\nu_1(1+z)}^{\nu_2(1+z)} d\nu \frac{d^2 L_x}{d\nu\, dV} \equiv \alpha\left[\nu_1(1+z), \nu_2(1+z), T_{gas}\right] n_e^2 \quad (3.6.2)$$

熱的 thermal brems の場合

$$\alpha_{ff}[\,] = \int_{\nu_1(1+z)}^{\nu_2(1+z)} d\nu \frac{2}{1+X} \frac{2^5 \pi e^6}{3 m_e c^3} \left(\frac{2\pi}{3 m_e k T_{gas}}\right)^{1/2} \bar{g}_{ff}(\nu, T_{gas}) \exp -\frac{h\nu}{kT_{gas}} \quad (3.6.3)$$

KOKUYO

No. 10

Date 97・7・9

$\bar{g}_{ff}(\nu, T_{gas}) \simeq 0.9 \left(\frac{h\nu}{kT_{gas}}\right)^{-0.3}$ (3.6.4)　　$\gamma(p,x) \equiv \int_0^x e^{-t} t^{p-1} dt$ 不完全ガンマ関数

$$\alpha_{ff} = \frac{1.8}{1+X} \cdot \frac{2^5 \pi e^6}{3 m_e c^3 h} \left(\frac{2\pi k T_{gas}}{3 m_e}\right)^{1/2} \left[\gamma\left(0.7, \frac{h\nu_2(1+z)}{kT_{gas}}\right) - \gamma\left(0.7, \frac{h\nu_1(1+z)}{kT_{gas}}\right)\right] \quad (3.6.5)$$

β-model を仮定

$$S_x(\theta, \nu_1, \nu_2) = \frac{\alpha[\nu_1(1+z), \nu_2(1+z), T_{gas}]}{4\pi (1+z)^4} \int_{-\infty}^{\infty} dl\, n_e^2 = \frac{\alpha[\,]}{4\pi(1+z)^4} \frac{\Gamma(3\beta - 1/2)}{\Gamma(3\beta)} n_0^2 r_0 \left[1 + \left(\frac{\theta}{r_0/d_A}\right)^2\right]^{-3\beta + \frac{1}{2}} \quad (3.6.8)$$

(ii) $(\Delta T/T)_{SZ}$: kinematic effect をみる

$$\left(\frac{\Delta T}{T_{CMB}}\right)_{SZ} = t(x)\, y(\theta) = t(x)\sqrt{\pi} \frac{\Gamma(\frac{3}{2}\beta - \frac{1}{2})}{\Gamma(\frac{3}{2}\beta)} \left(\frac{kT_{gas}}{m_e c^2}\right) \sigma_T n_0 r_0 \left[1 + \left(\frac{\theta}{r_0/d_A}\right)^2\right]^{-\frac{3}{2}\beta + \frac{1}{2}} \quad (3.6.9)$$

($t(x)$ ← (3.2.10)　$y(\theta)$ ← (3.3.3))

$S_x \propto n_0^2 r_0$ ┐
$(\Delta T/T)_{SZ} \propto n_0 r_0$ ┘ n_0 を消去して r_0 を求める

$$\Rightarrow r_0 = \frac{\alpha[\nu_1(1+z), \nu_2(1+z), T_{gas}]}{4\pi^{3/2}(1+z)^4 \sigma_T^2} \left\{t(x)\right\}^2 \left(\frac{m_e c^2}{kT_{gas}}\right)^2 \frac{\Gamma(\frac{3}{2}\beta)\Gamma(\frac{3}{2}\beta - \frac{1}{2})}{\Gamma(3\beta)\Gamma(\frac{3}{2}\beta - \frac{1}{2})} \left[1 + \left(\frac{\theta}{r_0/d_A}\right)^2\right]^{-\frac{1}{2}} \times \left.\frac{[\Delta T/T_{CMB}(\theta)]^2}{S_x(\theta, \nu_1, \nu_2)}\right|_{obs} \quad (3.6.10)$$

$S_x(\theta)$ の fitting ⇒ θ_0 = obs = $r_0/d_A(z)$ } $d_A(z)$

spherical symetric?
isothermal
inhomogenety
kinematics

§4 Spherical nonlinear model and the Press-Schechter theory

4.1 Theoretical counterpart of observed cluster of galaxies

"Theoretically" defined objects
optically
X-ray
} 同一である仮定

宇宙の銀河分布から予想される質量のゆらぎ

$\left\langle \left(\frac{\delta M}{M}\right)^2 \right\rangle^{1/2} \simeq 1$ at $r = 8 h^{-1}$ Mpc　　ゆらぎが非線形 ⇒ 重力収縮

$M(< 8h^{-1}\text{Mpc}) = \frac{4\pi}{3}(8h^{-1}\text{Mpc})^3 \times \bar{\rho}_0 \sim 6\times10^{14}\, \Omega_0 h^{-1} M_\odot$ typical cluster mass

$\Omega_0 \times \rho_{cr} = 2.8\times10^{11} h^2 M_\odot / \text{Mpc}^3$

4.2 Motion of Lagrangian mass shell

球対称 半径 r 内の mass M　　$\frac{d^2 r}{dt^2} = -\frac{GM}{r^2}$ (4.2.1)

KOKUYO

$\left(\frac{dr}{dt}\right)^2 = \frac{2GM}{r} - c$ (4.2.2)　　$c>0$ collapse　　P.J.E. Peebles "Large-scale structure of the Universe" (Princeton 1980)

$c<0$ expand

$dt = dr\sqrt{\frac{r}{2GM-cr}}$ (4.2.3)　　$2GM - cr = 2GM\cos^2\frac{\theta}{2}$ とおく

$r = \frac{2GM}{c}\sin^2\frac{\theta}{2} = \frac{GM}{c}(1-\cos\theta)$ (4.2.4)

$t = \int_0^\theta dt = \frac{GM}{c^{3/2}}(\theta - \sin\theta)$ (4.2.5)

(i) initial behavior ($\theta \ll 1$)

$r = \frac{GM}{2c}\theta^2\left(1-\frac{\theta^2}{12}\right)$ (4.2.6)

$t = \frac{GM}{6c^{3/2}}\theta^3\left(1-\frac{\theta^2}{20}\right)$ (4.2.7)

$\Rightarrow$

$r = \left[\frac{9}{2}GMt^2\right]^{1/3}\left[1-\frac{c}{20}\left(\frac{6t}{GM}\right)^{2/3}\right]$ (4.2.8)

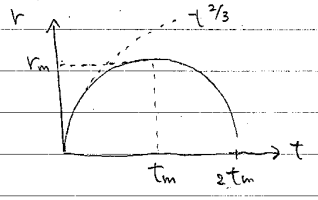
半径 $r(M,t)$ 内の平均質量密度

$\bar{\rho}(<r,t) = \frac{M}{\frac{4\pi}{3}r^3(M,t)} = \frac{1}{6\pi G t^2}\left[1+\frac{3c}{20}\left(\frac{6t}{GM}\right)^{2/3}\right]$ (4.2.9)

$= \rho_{EdS}(t)\left[1+\delta(<r,t)\right]$

$\delta(<r,t) \simeq \frac{3c}{20}\left(\frac{6t}{GM}\right)^{2/3}$ (4.2.10)

(ii) turn-arround (maximum expansion)



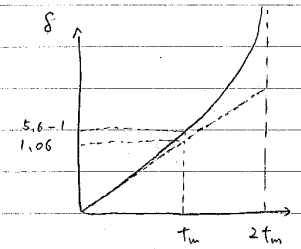
$\theta = \pi$ の時に max　　$t_m = \frac{\pi GM}{c^{3/2}}$, $r_m = \frac{2GM}{c}$ (4.2.11)

$\frac{\bar{\rho}(<r_m,t_m)}{\bar{\rho}_{EdS}(t_m)} = \frac{3M/4\pi r_m^3}{1/6\pi G t_m^2} = \frac{9\pi^2}{16} \simeq 5.6$ (4.2.12)

$\Rightarrow$ EdS ($\Omega_0 = 1, \lambda_0 = 0$) の場合

linear theory で extrapolate したやうな

$\delta_{linear}(<r;t_m) = \frac{3c}{20}\left(\frac{6t_m}{GM}\right)^{2/3} = \frac{3}{20}(6\pi)^{2/3} = 1.06$ (4.2.13)



(iii) virialization (≃ formation of objects)

$\theta = 2\pi$, $t_v = 2t_m$, $r = 0$　virial 平衡になる

turn around　virialize　dissipateless collapse

$0 - \alpha\frac{GM^2}{r_m} = K - \Omega = -\alpha\frac{GM^2}{2r_v}$　　$r_v = \frac{r_m}{2} = \frac{GM}{c}$ (4.2.14)

$\Omega = \alpha\frac{GM}{r_v}$

$\rho = -$定　$\alpha = 3/5$

$\rho = r^{-2}$　$\alpha = 1$

この時点での平均密度 $\frac{\bar{\rho}(<r_v,t_v)}{\bar{\rho}_{EdS}(t_v)} = \frac{8}{1/4}\times\frac{9\pi^2}{16} = 18\pi^2 \simeq 178$ (4.2.15)　Δ_{vir}

$\delta_c = \delta_{linear}(<r,t_v) = \frac{3}{20}(12\pi)^{2/3} \simeq 1.69$ (4.2.16)

COLLEGE NOTES / n028　　Source p.11 / 10 of 15 selected pages

No. 12

Date 97. 7. 9

$\Delta_{vir} \equiv \frac{\rho_{vir}(z_f)}{\bar{\rho}(z_f)} = 18\pi^2 f(\Omega_0, \lambda_0, z_f^{"z_v"})$

$\bar{\rho}_{EdS}(z_f)$

① z_f にできた object の密度 $\rho_{vir}(z_f) = \underbrace{18\pi^2 \times \Omega_0 \times \frac{3H_0^2}{8\pi G}}_{5\times10^{13}\,\Omega_0 h^{-1} M_\odot/\mathrm{Mpc}^3} (1+z_f)^3 \left[\frac{\Delta_{vir}(z_f)}{18\pi^2}\right]$ (4.2.17)

② z_f にできた M の object の virial 半径

$r_{vir}(z_f) \equiv \left(\frac{3M}{4\pi\rho_{vir}}\right)^{1/3} = 1.7 \left(\frac{\Delta_{vir}}{18\pi^2}\right)^{1/3} \left(\frac{M}{10^{15}M_\odot}\right)^{1/3} (\Omega_0 h^2)^{-1/3} (1+z_f)^{-1}$ Mpc (4.2.18)

③ z_f にできた object の温度

$T_{vir}(z_f) \equiv \frac{\mu m_p G M}{3k r_{vir}(z_f)} = 8.9 \left(\frac{1\mathrm{Mpc}}{r_{vir}(z_f)}\right)\left(\frac{M}{10^{15}M_\odot}\right)$ keV $= 5.3 \left(\frac{\Delta_{vir}}{18\pi^2}\right)^{1/3} \left(\frac{M}{10^{15}M_\odot}\right)^{2/3} (\Omega_0 h^2)^{1/3} (1+z_f)$ keV (4.2.19)

$\Rightarrow L_X(z_f, M) \propto T_{vir}^{1/2} \rho_{vir}^2 r_{vir}^3 \propto M^{4/3} \propto T_{vir}^2$

4.3 Press-Schechter theory

refs W. H. Press & P. Schechter ApJ 187 (1974) 425
R. Bower MNRAS 248 (1991) 332
Bond et al ApJ 379 (1991) 440

質量 $M \sim M+dM$ をもつ object の個数密度

$n_M(M)dM$ [コ/Mpc^3] mass function

temperature function $n_T(T) = n_M(M)\frac{dM}{dT}$ ← $M = M(T)$ ゆらぎの振幅 $\sigma_8 = \left\langle \left(\frac{\delta M}{M}\right)^2 \right\rangle^{1/2}_{r=8h^{-1}\mathrm{Mpc}}$

$n_L(L) = n_M(M)\frac{dM}{dL}$ ← $M = M(L)$

obs theory

4.4 PS mass function

(i) 密度ゆらぎの確率分布関数 $\delta(M, z_i)$: $z = z_i$ の時期に質量 M で平均したときの $\frac{\rho(M,z_i)}{\bar{\rho}(z_i)} - 1$ の値

$\vec{r}$ のまわりに半径 r_M の球をとって その mass を求める $r_H \equiv \left(\frac{3M}{4\pi\rho_0}\right)^{1/3}(1+z)^{-1}$

現在の宇宙の mean density

comoving length

$\delta(M, z_i; \vec{r}) \equiv \int d\vec{r}'\, W(\vec{r}'-\vec{r}; r_M) \frac{\rho(\vec{r}', z_i) - \bar{\rho}(z_i)}{\bar{\rho}(z_i)}$ (4.4.1)

$\int d\vec{x}\, W(\vec{x}; r_M) = 1$ (4.4.2)

Heaviside function

(A) top-hat $W_{TH}(\vec{x}; r_M) = \frac{3}{4\pi r_M^3}\theta(R_H - |\vec{x}|)$ $M_{TH} = \int_0^{R_{TH}} \bar{\rho}\, 4\pi r^2 dr = \frac{4\pi}{3} R_{TH}^3 \bar{\rho}$ $R_{TH} = \left(\frac{3M_{TH}}{4\pi\bar{\rho}}\right)^{1/3}$

$\tilde{W}_{TH}(\vec{k}; R_M) = \frac{3}{k^3 R_M^3}\left[\sin(kR_M) - kR_M\cos(kR_M)\right]$

KOKUYO

No. 13

Date 97・7・9

(B) Gaussian $W_G(|\vec{x}|;R_M) = \frac{1}{[2\pi R_M^2]^{3/2}} \exp\left[-\frac{x^2}{2R_M^2}\right]$

$M_G = \int_0^\infty \bar{\rho} \exp\left(-\frac{r^2}{2R_M^2}\right) 4\pi r^2 dr = (2\pi)^{3/2} \bar{\rho} R_M^3 \quad \therefore R_M = \frac{1}{\sqrt{2\pi}}\left(\frac{M_G}{\bar{\rho}}\right)$

$\tilde{W}_G(k,R_M) = \exp\left(-\frac{k^2 R_M^2}{2}\right)$

⊙ random-Gaussian density field

⇓

δ の分布関数が $P(\delta) = \frac{1}{\sqrt{2\pi\sigma_M^2(z_i)}} \exp\left[-\frac{\delta^2}{2\sigma_M^2(z_i)}\right]$ (4.4.3)

$\sigma_M^2(z_i) \equiv \langle \{\delta(M,z_i)\}^2 \rangle = \frac{1}{(2\pi)^3} \int d\vec{k}\, \tilde{W}^2(\vec{k};r_M) \underbrace{\langle |\delta(\vec{k};z_i)|^2 \rangle}_{P(k,z_i)}$ (4.4.4)

(ii) Probability of collapsed objects

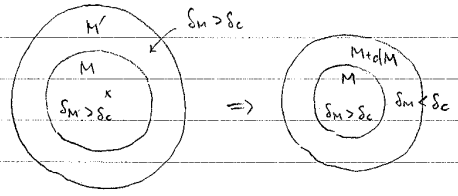
Spherical nonlinear model

ゆらぎの成長を線形理論で記述する。 linear growth rate $D(z) = (1+z)^{-1}$

$\delta_{linear}(M,z) \equiv \frac{D(z)}{D(z_i)} \underbrace{\delta(M,z_i)}_{z_i \to \infty}$ (4.4.5)

ある epoch z において、ゆらぎの値が δ_c をこえている割合

$P[\delta_{linear}(M,z) > \delta_c] = P[\underbrace{\delta(M,z_i)}_{\delta_{linear}(M,z_i)} > \underbrace{\frac{D(z_i)}{D(z)}\delta_c}_{\delta_{c,i}}] = \int_{\delta_{ci}}^\infty \frac{1}{\sqrt{2\pi\sigma_M^2(z_i)}} \exp\left[-\frac{\delta^2}{2\sigma_M^2(z_i)}\right] d\delta$ (4.4.6)



⊙ z_i において M が独立した object である確率

M がより大きな object の一部ではないことを保証する

$P(M,z,\delta_c)dM = P[\delta_{linear}(M,z) > \delta_c] \times P[\delta_{linear}(M+dM,z) < \delta_c \mid \delta_{linear}(M,z) > \delta_c]$

$= \{1 - P[\delta_{linear}(M+dM,z) > \delta_c \mid \delta_{linear}(M,z) > \delta_c]\}$

$= P[\delta_{linear}(M,z) > \delta_c] - P[\delta_{linear}(M+dM,z) > \delta_c]$

$= -\frac{\partial}{\partial M} P[\delta_{linear}(M,z) > \delta_c]\, dM$ (4.4.7)

(iii) $\partial P/\partial M$ to PS mass function

(4.4.6) $\Rightarrow P[\delta_{linear}(M,z) > \delta_c] = \frac{1}{\sqrt{2\pi}} \int_{\underbrace{\delta_{ci}/\sigma_M(z_i)}_{\delta_c/\sigma_M(z)}}^\infty e^{-\frac{x^2}{2}} dx$ (4.4.8)

↙ linear theory で extrapolate したもの。z_i によらない

$\frac{\partial P}{\partial M} = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{\delta_c^2}{2\sigma_M^2(z)}\right] \frac{d}{dM}\left[\frac{\delta_c}{\sigma_M(z)}\right]$

$= \frac{1}{\sqrt{2\pi}} \frac{\delta_c}{\sigma_M^2(z)} \frac{d\sigma_M(z)}{dM} \exp\left[-\frac{\delta_c^2}{2\sigma_M^2(z)}\right]$ (4.4.9)

0

KOKUYO

COLLEGE NOTES / n028

Source p.13 / 12 of 15 selected pages

No. 14

Date 97・7・9

$$n(M,z) = \frac{\bar{\rho}_0}{M} \times p(M,z,\delta_c) = \frac{\bar{\rho}_0}{\sqrt{2\pi}} \frac{\delta_c}{M} \frac{1}{\sigma_M^2(z)} \left| \frac{d\sigma_M(z)}{dM} \right| \exp\left[-\frac{\delta_c^2}{2\sigma_M^2(z)} \right] \quad (4.4.10)$$

$$\int_0^\infty dM\, M n(M,z) = \bar{\rho}_0 \int_0^\infty \left(-\frac{\partial P}{\partial M}\right) dM = \bar{\rho}_0 \left[P_{M=0} - P_{M=\infty} \right] = \bar{\rho}_0 \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-\frac{x^2}{2}} dx = \frac{\bar{\rho}_0}{2} \quad (4.4.11)$$

($P_{M=\infty} = 0$)

$$n_{PS} = 2 \times n \quad (4.4.12)$$

§5 Statistics of clusters of galaxies

@ spatial correlation

銀河 $\xi_{gg} = (r/r_{0,gg})^{-1.8}$ $r_{0,gg} \simeq 5.4\, h^{-1}$ Mpc Totsuji & Kihara (1969)

銀河団

$\xi_{cc,i} \simeq (r/0.4 d_i)^{-1.8}$ $d_i = n_i^{-\frac{1}{3}}$ mean separation n_i 個数密度 (コ/Mpc^3) i : カタログのインデックス

$\xi_{cc,i} = A_i r^{-1.8}$ ($= (0.4 d_i)^{1.8}$) 銀河の bias モデル Kaiser (1984?)

@ optical luminosity function

銀河 Schechter func $\phi(L_{opt})\, dL = \phi^* \left(\frac{L_{opt}}{L_*}\right)^{-\alpha} \exp\left[-\left(\frac{L_{opt}}{L_*}\right)\right] d\left(\frac{L_{opt}}{L_*}\right)$

ϕ^*	$\sim 0.02\, h^3$ Mpc^3	$1.3\times10^{-5} h^3$ Mpc^{-3}
L^*	$\sim 7\times10^9\, h^{-2} L_\odot$	$2.5\times10^{12} h^{-2} L_\odot$
α	~ 1	2

B-band	銀河	cluster
mean separation	$3\, h^{-1}$ Mpc	$50\, h^{-1}$ Mpc
local $(100\,\text{Mpc})^3$	3万コ	8コ

@ mass function

obs : むずかしい

銀河の mass (中心から半径 … まで)

$M_A \equiv M(< 1.5\, h^{-1}$ Mpc)

emprical $M_A - L_{opt}$

M_A – velocity dispersion

cumulative

$$\Rightarrow n(>M_A) = (4 \pm 0.6) \times 10^{-5} \left(\frac{M_A}{M_*}\right)^{-1} \exp\left(-\frac{M_A}{M_*}\right) h^3 \text{Mpc}^{-3}$$

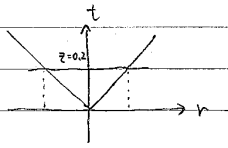
$M_* = (1.8 \pm 0.3) \times 10^{14} h^{-1} M_\odot$

@ X-ray luminosity Function

$n(L, z=0)$

$n(T, z=0)$

$n(M, z=0)$



ROSAT Brightest Cluster Sample BCS

$z \le 0.3$ ($z > \sim 0.1$)

$F \to L$ ($h = 0.5$, $q_0 = 1/2$)

$$\phi(L_x) = A L_x^{-\alpha} \exp\left(-\frac{L_x}{L_*}\right)$$

$A = (4.6 \pm 0.8)\, 10^{-6}\, h^3\, \text{Mpc}^{-3} \left[0.25 \times 10^{44} h^{-2} \text{erg/s}\right]^{\alpha-1}$

$L_* = (2.2 \pm 0.4) \times 10^{44} h^{-2}$ erg/s

$\alpha = 1.8 \pm 0.1$

KOKUYO

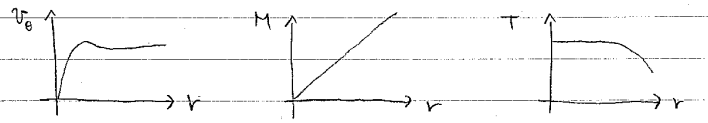
COLLEGE NOTES / n028

Source p.14 / 13 of 15 selected pages

No. 15

Date 97 · 7 · 10

@ X-ray temperature function



$n(T) = 1.2^{+2.6}_{-0.9} \times 10^{-3}\ h^3\ Mpc^{-3}\ keV^{-1}\ (T/1keV)^{-5.1\pm0.7}$ Henry et al (1997)

@ Temperature - luminosity relation

$$L_{bol} = L_{44}\left(\frac{T}{6keV}\right)^{\alpha} \times 10^{44}\ h^{-2}\ erg/s \qquad \begin{cases} L_{44} = 2.9^{+2.6}_{-1.4} \\ \alpha = 3.4 \end{cases}$$

@ Cluster abundances $\iff (\Omega_0, \sigma_8)$

obs : $n(M)$, $n(L)$, $n(T)$, $N(>S)$
⇕ $n(m)$

theory : simulation : reliability, parameter search など

↳ PS theory : "definition", spherical collapse, ···

internal check

@ $P(k)$ からズレの スペクトル ⇒ P.S. theory の予言

@ Dynamical evolution : (Ω_0, λ_0)

CDM の予言 fitting formula

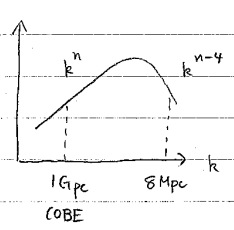
$$T(k) = \left[\frac{\ln(1+2.34q)}{2.34q}\right]^2 \left[1 + 3.89q + (16.18)^2 + (5.46q)^3 + (6.71q)^4\right]^{-\frac{1}{2}}$$

$q = k/\Gamma(h^{-1}Mpc)^{-1}$

$0.25 \to \Gamma = \Omega_0 h \exp\left[-\Omega_b\left(1+\frac{\sqrt{2h}}{\Omega_0}\right)\right]$

$$P(k,z) = A\left[\frac{D(z)}{D(z_i)}\right]^2 k^n T^2(k)$$

↓ σ_8 $\quad n=1$: HZ



$\xi_{gg}(r) = b^2\ \xi_{3DM}(r)$

↓ bias ↓

$\sigma_{8,gg} = 1$ $\qquad$ $\sigma_{8,DM}$

$$b \equiv \frac{\sigma_{8,gg}}{\sigma_{8,DM}} = \frac{1}{\sigma_{8,DM}} \qquad \begin{matrix} \sigma_8 < 1 & b > 1 \\ \sigma_8 > 1 & b < 1 \end{matrix}$$

$b = 1.5$ ($\Omega_0 = 1$, CDM)

KOKUYO

No. 16

Date 97・7・10

レポート

A4 3枚 / 1人

1) X-ray : cluster の T, L, M, core radius 等、いづれか1つを観測的に決定する方法とその不定性を示せ

2) radio : SZ effect を obs する最適のバンドを示せ (noise 源は何か?)
$\Delta T_{SZ}/T_{CMB}$ はどこまで obs できるか

3) submm : SZ effect を obs する可能性について

4) optical : cluster of galaxies の σ_v, z を決定する方法と不定性

5) kinematic SZ の表式をきちんと導け

6) (Ω_0, λ_0) 一般の場合の δ_c, Δ_{vir} を計算し Eke etal の図を再現せよ

〆切 9月16日 までに A棟事務

KOKUYO